Ex 1. Se dă următorul tabel de adevăr:

1. Să se minimizeze funcția. (1p)
2. Să se implementeze circuitul cu porți NAND (2p)
3. Să se implementeze circuitul cu MUX 4:1 în cascadă. Pe nivelul din stg. → var. sel. b și d dr → var. sel. c și a (4p)
4. 1 singur MUX 4:1 cu var a și d (4p)

a	b	c	d	f
0	0	0	0	X
0	0	0	1	X
0	0	1	0	0
0	0	1	1	*
0	1	0	0	Y
0	1	0	1	1
0	1	1	0	1
0	1	1	1	X
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	Y
1	1	0	0	X
1	1	0	1	Y
1	1	1	0	1
1	1	1	1	*

fiecare grup o scriu din diagramă

REZOLVARE

1.

ab \ cd	00	01	11	10
00	X	X	*	0
01	Y	1	X	1
11	X	Y	*	1
10	1	1	Y	0

Pas 1. Minimizare pt. 1

ab \ cd	00	01	11	10
00	0	0	*	0
01	0	1	0	1
11	0	0	*	1
10	1	1	0	0

$f_1 = a\bar{b}\bar{c} + bc\bar{d} + \bar{a}b\bar{c}d$

Pas 2. Minimiz. pt X

ab \ cd	00	01	11	10
00	1	1	*	0
01	0	*	1	*
11	1	0	*	*
10	*	*	0	0

$f_2 = X(\bar{b}\bar{c} + bc + a\bar{c}\bar{d}) =$
$= X\bar{b}\bar{c} + Xbc + Xa\bar{c}\bar{d}$

Minimiz pt. Y

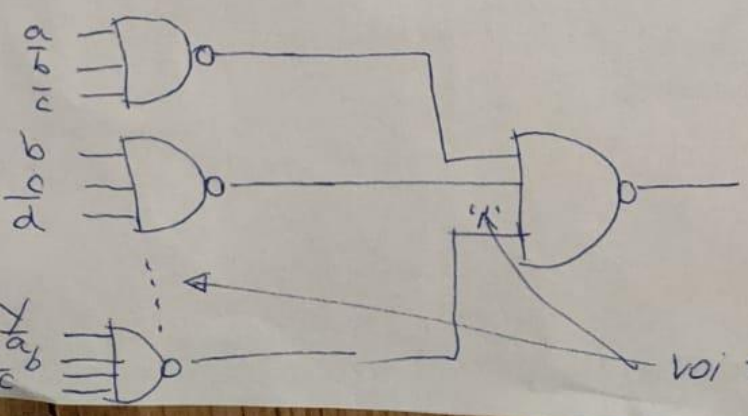
ab \ cd	00	01	11	10
00	0	0	*	0
01	1	*	0	*
11	0	1	*	*
10	*	*	1	0

$f_3 = Y(ad + \bar{a}b\bar{c}) =$
$= Yad + Y\bar{a}b\bar{c}$

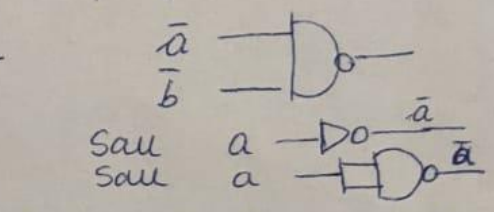
$f = f_1 + f_2 + f_3 = a\bar{b}\bar{c} + bc\bar{d} + \bar{a}b\bar{c}d + X\bar{b}\bar{c} + Xbc + Xa\bar{c}\bar{d} + Yad + Y\bar{a}b\bar{c}$

$f = \bar{\bar{f}} = \overline{\bar{a}\bar{b}\bar{c} + bc\bar{d} + \bar{a}b\bar{c}d + X\bar{b}\bar{c} + Xbc + Xa\bar{c}\bar{d} + Yad + Y\bar{a}b\bar{c}}$

$= \overline{a\bar{b}\bar{c}} \cdot \overline{bc\bar{d}} \cdot \overline{\bar{a}b\bar{c}d} \cdot \overline{X\bar{b}\bar{c}} \cdot \overline{Xbc} \cdot \overline{Xa\bar{c}\bar{d}} \cdot \overline{Yad} \cdot \overline{Y\bar{a}b\bar{c}}$



Dacă aveți intrări variabile negate, puteți să le scrieți direct așa!

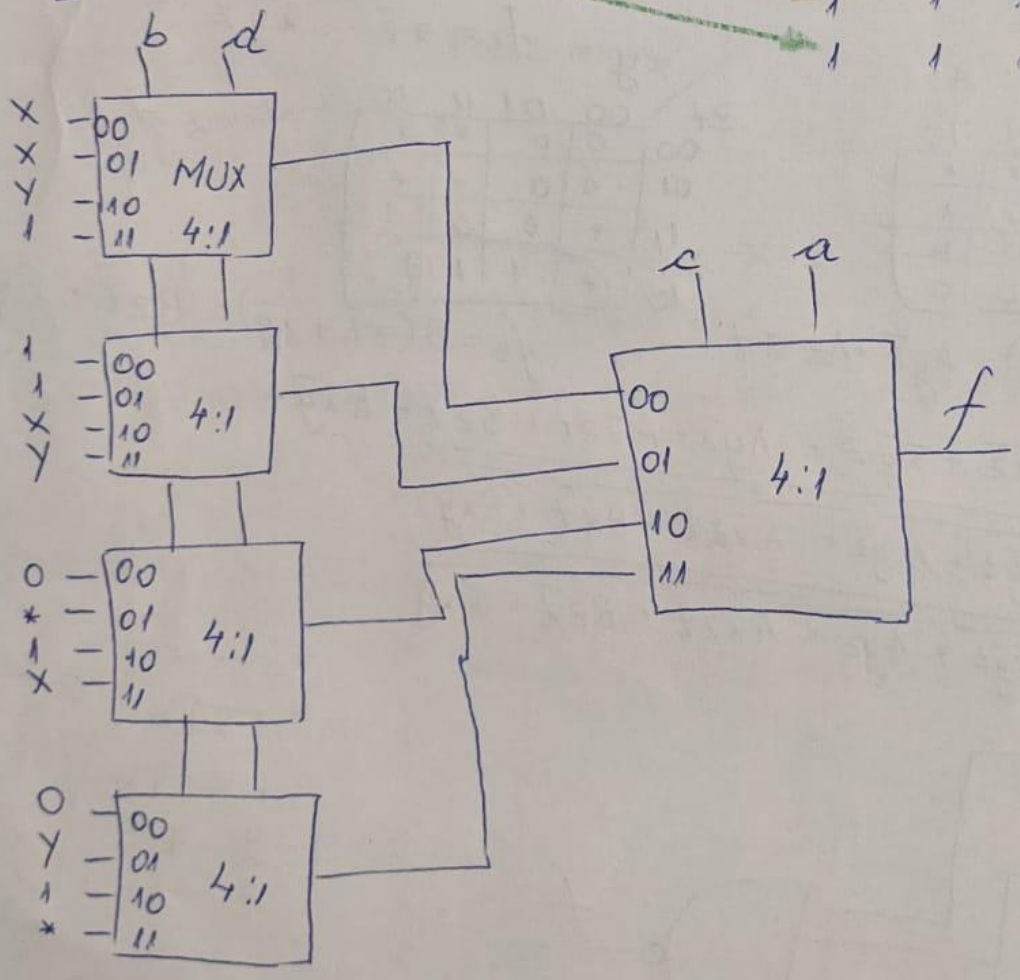


voi trebuie să desenați toate porțile!!

(2)

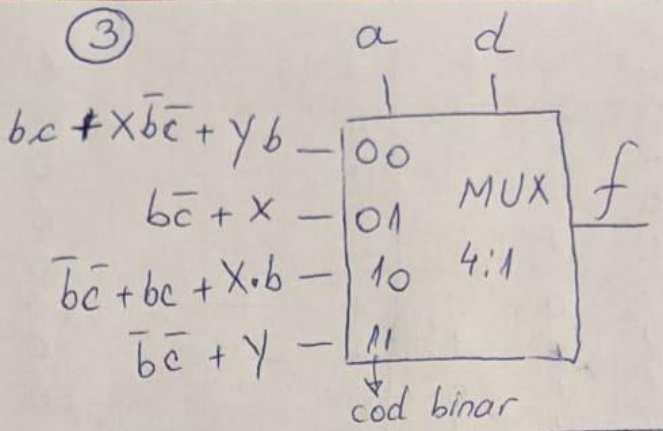
a	b	c	d	f
0	0	0	0	X
0	0	0	1	X
0	0	1	0	0
0	0	1	1	*
0	1	0	0	Y
0	1	0	1	1
0	1	1	0	1
0	1	1	1	X
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	Y
1	1	0	0	X
1	1	0	1	Y
1	1	1	0	1
1	1	1	1	*

c	a	b	d	f
0	0	0	0	X
0	0	0	1	X
0	0	1	0	Y
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	X
0	1	1	1	Y
1	0	0	0	0
1	0	0	1	*
1	0	1	0	1
1	0	1	1	X
1	1	0	0	0
1	1	0	1	Y
1	1	1	0	1
1	1	1	1	*



ab \ cd (cod Grey)	00	01	11	10
00	x	x	*	0
01	y	1	x	1
11	x	y	*	1
10	1	1	y	0

③



I $\boxed{ab = 00}$

b \ c	0	1
0	x	0
1	y	1

Pas 1 minimiz. pt 1

b \ c	0	1
0	0	0
1	0	1

Pas 2 - minimiz pt x

	1	0
	0	*

pt y

	0	0
	1	*

$f = bc + x\bar{b}\bar{c} + yb$

II $\boxed{ab = 01}$

b \ c	0	1
0	x	*
1	1	x

Pas 1. minimiz. pt 1

b \ c	0	1
0	0	*
1	1	0

Pas 2 minimiz. pt x:

b \ c	0	1
0	1	*
1	*	1

$f = b\bar{c} + x \cdot 1$
$= b\bar{c} + x$

III $\boxed{ab = 10}$

b \ c	0	1
0	1	0
1	x	1

1:

b \ c	0	1
0	1	0
1	0	1

x:

b \ c	0	1
0	*	0
1	1	*

$f = \bar{b}\bar{c} + bc + x \cdot b$

IV $\boxed{ab = 11}$

b \ c	0	1
0	1	y
1	y	*

1:

b \ c	0	1
0	1	0
1	0	*

y:

	*	1
	1	*

$f = \bar{b}\bar{c} + y$

Ex 2 ① Minimizare și implementare porți NAND. ② Implem. MUX 4:1 – nivelul stg t, y – nivelul dr. x, z

z	y	x	t	f
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	*
0	1	0	0	A
0	1	0	1	A
0	1	1	0	B
0	1	1	1	A
1	0	0	0	*
1	0	0	1	*
1	0	1	0	B
1	0	1	1	B
1	1	0	0	1
1	1	0	1	1
1	1	1	0	B
1	1	1	1	0

fiecare grupe o coloana din diagramă

① Zt \ xy

Zt	00	01	11	10
00	0	A	1	*
01	0	A	1	*
11	*	A	0	B
10	1	B	B	B

③ 1 singur MUX 4:1 cu var y, z, t

Pasul 1. Minimizare pt. 1

Zt \ xy	00	01	11	10
00	0	0	1	*
01	0	0	1	*
11	*	0	0	0
10	1	0	0	0

$f_1 = x\bar{z} + \bar{x}\bar{y}z$

Pasul 2. Minimiz pt. A

Zt \ xy	00	01	11	10
00	0	1	*	*
01	0	1	*	*
11	*	1	0	0
10	*	0	0	0

$f_2 = A(y\bar{z} + \bar{x}zt) = Ay\bar{z} + A\bar{x}zt$

Minimiz. pt. B

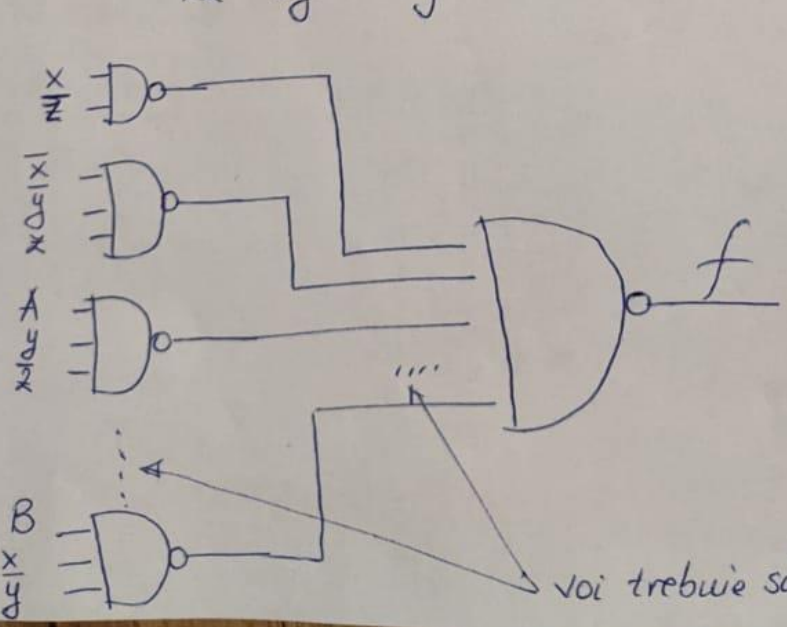
Zt \ xy	00	01	11	10
00	0	0	*	*
01	0	0	*	*
11	*	0	0	1
10	*	1	1	1

$f_3 = B(z\bar{t} + x\bar{y}) = Bz\bar{t} + Bx\bar{y}$

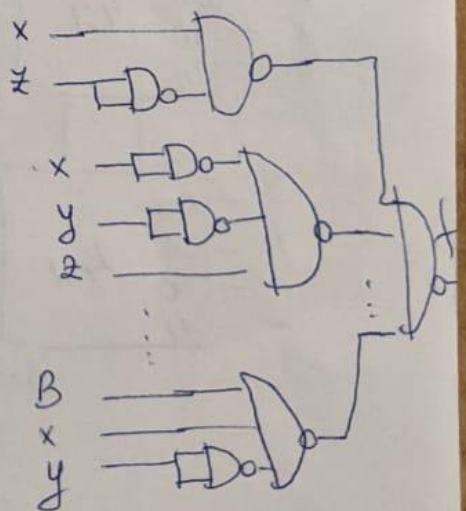
$f = f_1 + f_2 + f_3 = x\bar{z} + \bar{x}\bar{y}z + Ay\bar{z} + A\bar{x}zt + Bz\bar{t} + Bx\bar{y} =$

$= \bar{\bar{f}} = \overline{x\bar{z} + \bar{x}\bar{y}z + Ay\bar{z} + A\bar{x}zt + Bz\bar{t} + Bx\bar{y}} =$

$= \overline{\overline{x\bar{z}} \cdot \overline{\bar{x}\bar{y}z} \cdot \overline{Ay\bar{z}} \cdot \overline{A\bar{x}zt} \cdot \overline{Bz\bar{t}} \cdot \overline{Bx\bar{y}}}$



SAU

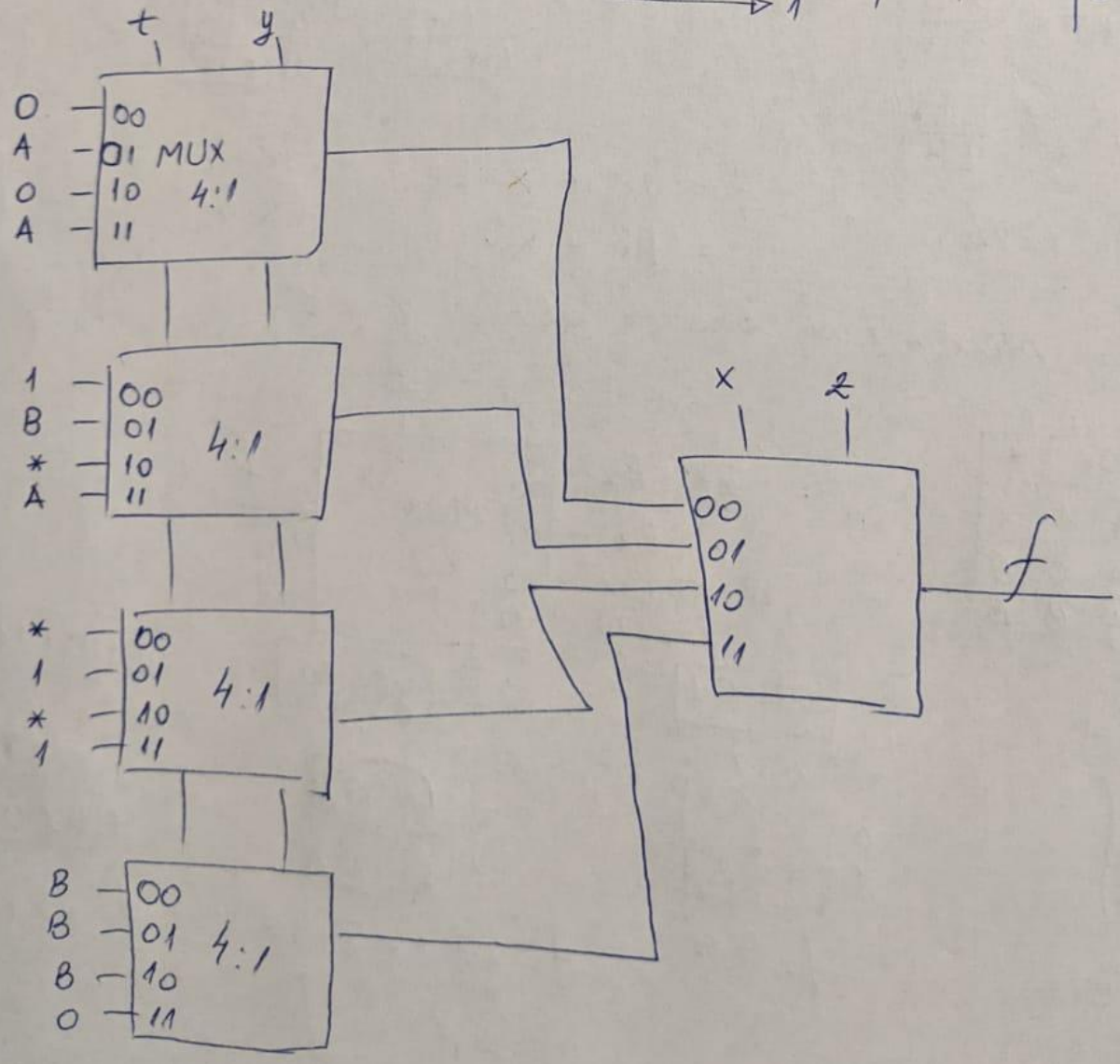


voi trebuie să desenați toate porțile !!

(2)

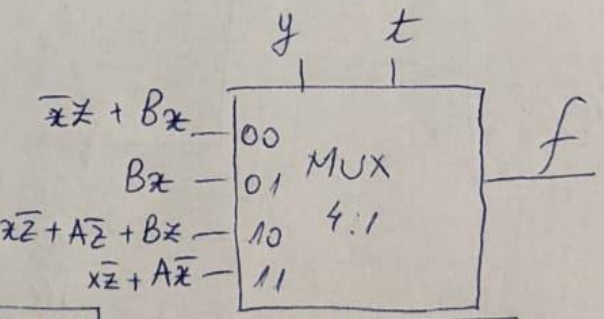
x	y	z	t	f
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	*
0	1	0	0	A
0	1	0	1	A
0	1	1	0	B
0	1	1	1	A
1	0	0	0	*
1	0	0	1	*
1	0	1	0	B
1	0	1	1	B
1	1	0	0	1
1	1	0	1	1
1	1	1	0	B
1	1	1	1	0

x	z	t	y	f
0	0	0	0	0
0	0	0	1	A
0	0	1	0	0
0	0	1	1	A
0	1	0	0	1
0	1	0	1	B
0	1	1	0	*
0	1	1	1	A
1	0	0	0	*
1	0	0	1	1
1	0	1	0	*
1	0	1	1	1
1	1	0	0	B
1	1	0	1	B
1	1	1	0	B
1	1	1	1	0



③ 1 singur MUX 4:1 cu var de selecție y și t

zt \ xy	00	01	11	10
00	0	A	1	*
01	0	A	1	*
11	*	A	0	B
10	1	B	B	B



I $yt = 00$

z \ x	0	1
0	0	*
1	1	B

$f = \bar{x}z + B \cdot x$

minimiz. pt. 1 — minimiz. pt B

II. $yt = 01$

z \ x	0	1
0	0	*
1	*	B

$f = B \cdot x$

III. $yt = 10$

z \ x	0	1
0	A	1
1	B	B

$f = x\bar{z} + A \cdot \bar{z} + B \cdot z$

minimiz. pt. 1 — minimiz pt A — minimiz. pt B

IV $yt = 11$

z \ x	0	1
0	A	1
1	A	0

$f = x\bar{z} + A \cdot \bar{x}$